

In the following exercises, evaluate each definite integral using the Fundamental Theorem of Calculus, Part 2.

$$\begin{aligned}
 170. \int_{-1}^2 (x^2 - 3x) dx &= \left[\frac{x^3}{3} - \frac{3}{2}x^2 \right]_{-1}^2 \\
 &= \left[\frac{(2)^3}{3} - \frac{3}{2}(2)^2 \right] - \left[\frac{(-1)^3}{3} - \frac{3}{2}(-1)^2 \right] \\
 &= \left[\frac{8}{3} - 6 \right] - \left[-\frac{1}{3} - \frac{3}{2} \right] \\
 &= \frac{8}{3} - 6 + \frac{1}{3} + \frac{3}{2} \\
 &= \frac{8}{3} + \frac{1}{3} - 6 + \frac{3}{2} \\
 &= 3 - 6 + \frac{3}{2} = -3 + \frac{3}{2} \\
 &= -\frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 171. \int_{-2}^3 (x^2 + 3x - 5) dx &= \left[\frac{x^3}{3} + \frac{3}{2}x^2 - 5x \right]_{-2}^3 \\
 &= \left[\frac{(3)^3}{3} + \frac{3}{2}(3)^2 - 5(3) \right] - \left[\frac{(-2)^3}{3} + \frac{3}{2}(-2)^2 - 5(-2) \right] \\
 &= \left[9 + \frac{27}{2} - 15 \right] - \left[-\frac{8}{3} + 6 + 10 \right] \\
 &= 9 - 15 + \frac{27}{2} + \frac{8}{3} - 6 - 10 \\
 &= 9 - 15 - 6 - 10 + \frac{27}{2} + \frac{8}{3} \\
 &= -22 + \frac{81}{6} + \frac{16}{6} = -22 + \frac{97}{6} = \frac{-132 + 97}{6} \\
 &= \frac{-35}{6}
 \end{aligned}$$

$$\begin{aligned}
 172. \int_{-2}^3 (t+2)(t-3) dt &= \int_{-2}^3 t^2 - t - 6 dt = \left[\frac{t^3}{3} - \frac{t^2}{2} - 6t \right]_{-2}^3 \\
 &= \left[\frac{(3)^3}{3} - \frac{(3)^2}{2} - 6(3) \right] - \left[\frac{(-2)^3}{3} - \frac{(-2)^2}{2} - 6(-2) \right] \\
 &= \left[9 - \frac{9}{2} - 18 \right] - \left[-\frac{8}{3} - 2 + 12 \right] \\
 &= 9 - 18 - \frac{9}{2} + \frac{8}{3} - 10 = -19 - \frac{27}{6} + \frac{16}{6} = -19 - \frac{11}{6} = \frac{-114 - 11}{6} \\
 &= \frac{-125}{6}
 \end{aligned}$$

$$\begin{aligned}
 182. \int_0^{2\pi} \cos \theta d\theta &= \left[\sin \theta \right]_0^{2\pi} \\
 &= \sin(2\pi) - \sin(0) \\
 &= 0 - 0 = 0
 \end{aligned}$$

$$\begin{aligned}
 183. \int_0^{\pi/2} \sin \theta d\theta &= -\cos \theta \Big|_0^{\pi/2} \\
 &= \left[-\cos\left(\frac{\pi}{2}\right) - (-\cos(0)) \right] \\
 &= [-0 + 1] = 1
 \end{aligned}$$

$$184. \int_0^{\pi/4} \sec^2 \theta d\theta = \left[\tan(\theta) \right]_0^{\pi/4} = \left[\tan\left(\frac{\pi}{4}\right) - \tan(0) \right]$$

$$= 1 - 0 = 1$$

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$187. \int_{\pi/4}^{\pi/2} \csc^2 \theta d\theta = \left[-\cot(\theta) \right]_{\pi/4}^{\pi/2} = \left[-\cot\left(\frac{\pi}{2}\right) - (-\cot\left(\frac{\pi}{4}\right)) \right]$$

$$= 0 + \cot\left(\frac{\pi}{4}\right) = 0 + 1 = 1$$

$$\cot(\theta) = \frac{\cos(\theta)}{\sin(\theta)}$$